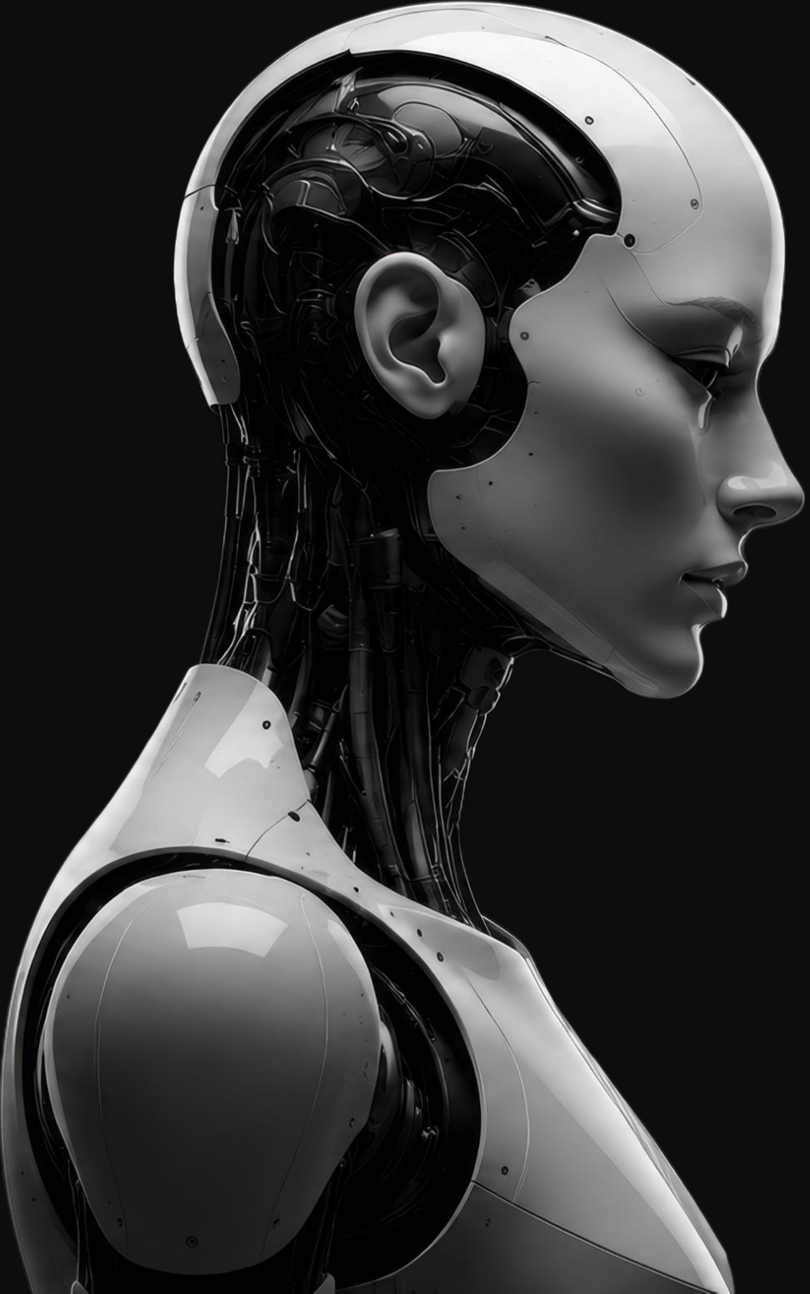


sequor robotics

Scaling Physical AI through Robot Foundation Models

2026 Investor Relations

2000

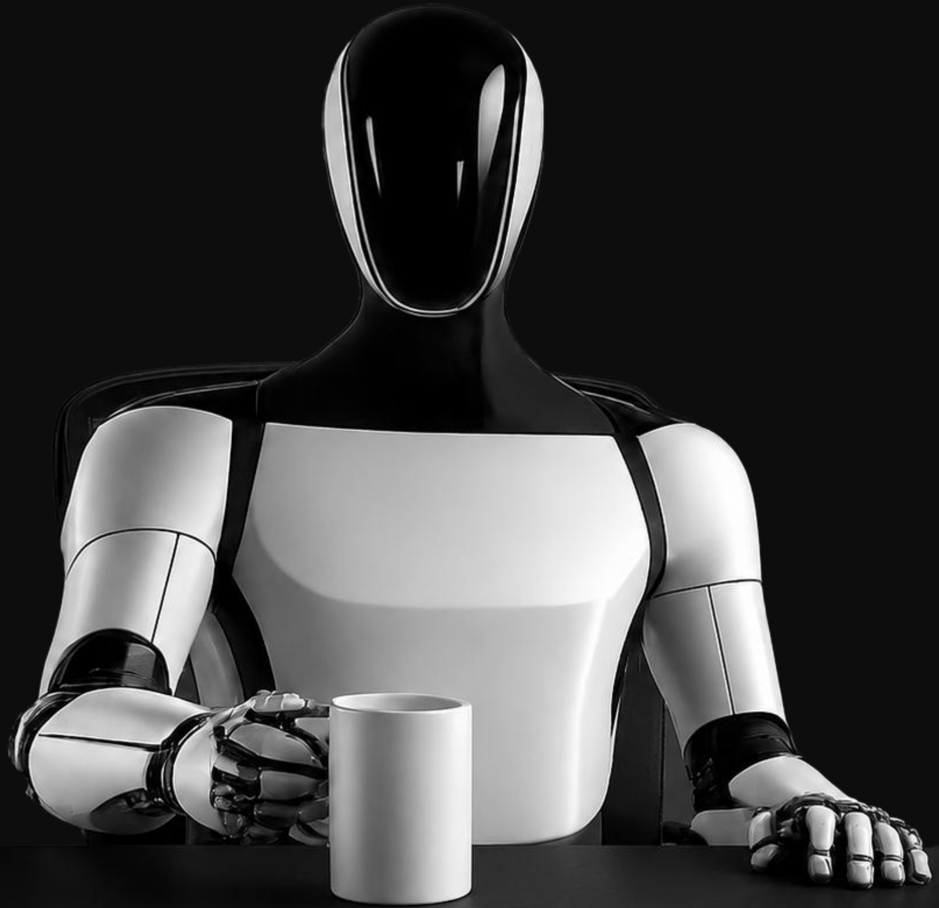
47.6

1

2000 Data

47.6 Hours

1 Task

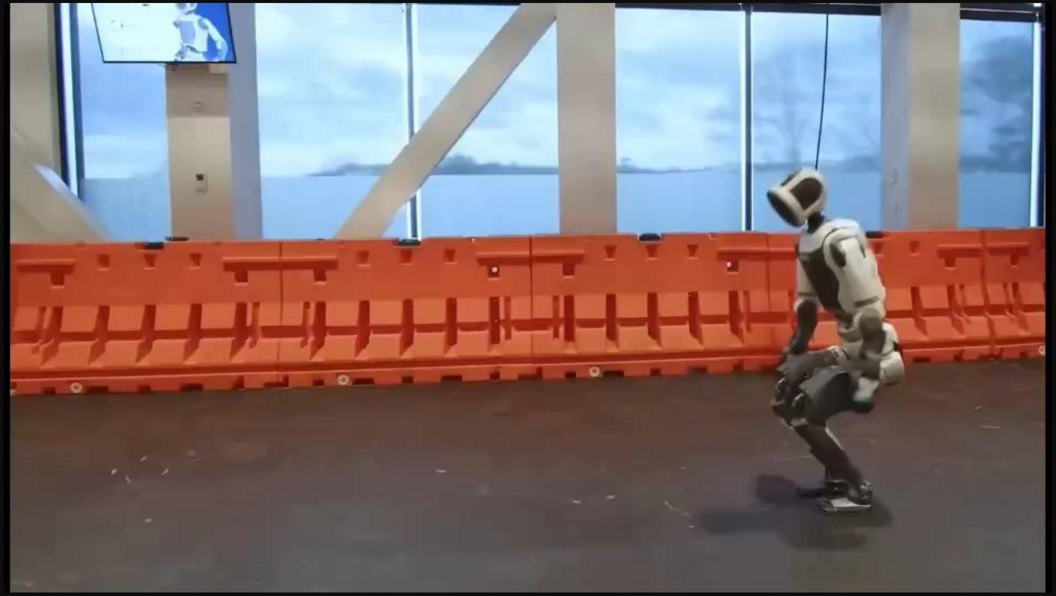


Training robots needs vast amounts of data and vast amounts of time.

Ten years have passed, but what robots can actually do in the real world hasn't changed much.

2016 Boston Dynamics Atlas Demo Video

2025 Boston Dynamics Atlas Demo Video



Sequor Robotics,
where data becomes robot intelligence

Beyond the limits of 2D,
we build and research 3D data
so robots can understand the world humans actually
live in — shifting the paradigm of robot intelligence.

Just as LLMs grew by training on massive amounts of data, robots too can only advance by learning from massive amounts of data.

2015

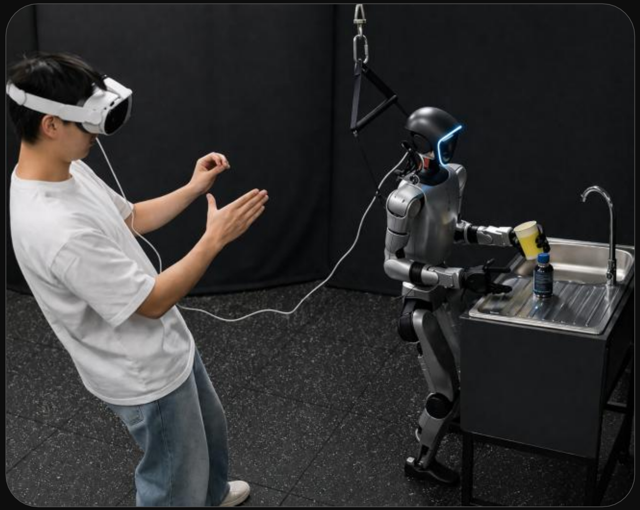
Rule-Based Systems

Operating through combinations of human-defined rules and modules

2021

Imitation Learning from Real Data

Collecting real-world data and training via teleoperation



2024

2025

The Limits of Real Data

Despite growing data volumes, generalization continues to fall short

Emerging approaches to robot data collection

sequor
robotics

Egocentric Video

Collecting manipulation data from the robot's perspective using a first-person camera mounted on the robot arm.

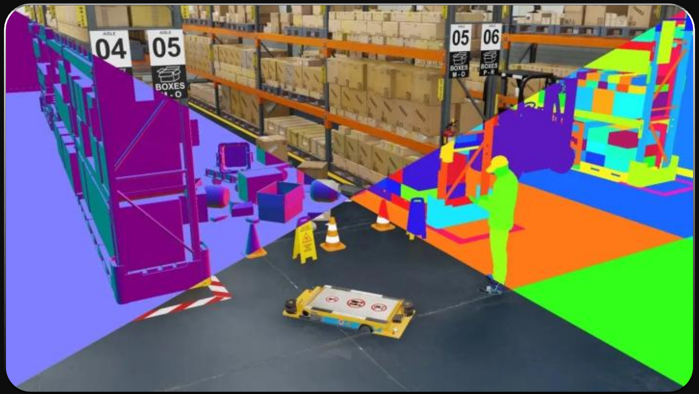
Scene Generation

Collecting large-scale, diverse datasets by automatically generating various environments and object placements.

Human motion capture

Capturing human motions directly to teach robots natural movement patterns.

Simulation



Rapidly generating large-scale training data in virtual environments without physical constraints.

Why Simulation?

Overwhelming Data Volume & Cost Efficiency

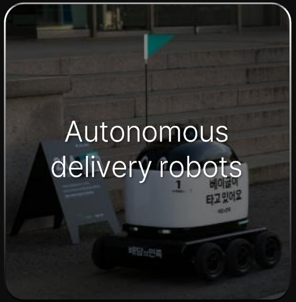
Hundreds to thousands of scenarios running in parallel, 24/7

Minimal human involvement, high margins

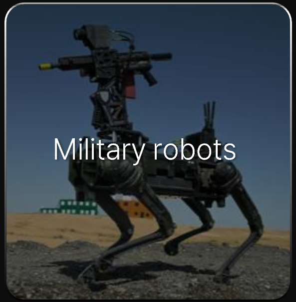
Method	Data Volume	Usability	Key Costs
Simulation	**Unlimited**	**Moderate (sim-to-real gap)**	• **GPU/compute infrastructure** • **Simulation environment build (one-time upfront cost)** • **Software licensing**
Motion Capture	Low	Easy	• MoCap system setup: multiple cameras, calibrated studio space • Data processing engineer labor
Egocentric Video	High	Hard	• Collection devices: head-mounted cameras, many participants • Labeling costs
Scene Generation	High	Hard (physics inaccuracy)	• GPU compute: training generative models and running large-scale inference • Generative model licensing / API fees
Teleoperation	Moderate	Easy	• Labor: skilled operators required for every hour of operation • Robot hardware • Teleoperation interface equipment

Every robotics company needs Simulation data

Safety Critical — High Risk Scenarios



on-road collision



real combat environments can't be recreated

fire and structural collapse sites

Industrial — Large Scale Data Generation

pick & place, assembly, hundreds of thousands of repetitions

unlimited SKU types, many environmental variables

diverse crops and weather conditions

sequor robotics

Human Interaction — Human Variables

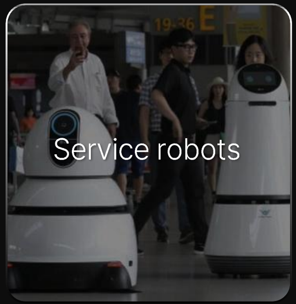
Diverse physical conditions

unpredictable behavior patterns

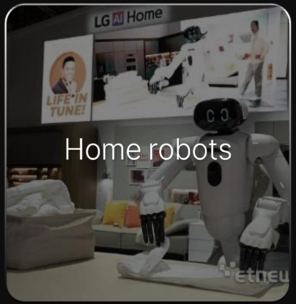
shared spaces with humans, safety required

General Intelligence — Complex Environments

generalized tasks



hotels, offices



different layouts in every home, multi-step tasks

Customer Pain Point

Even when simulation is needed, adoption is expensive and painful.

QA/QC Outsourcing Costs

Real robot damage during testing

Testing & debugging 60% of the job

Hardware delays
Compress development timelines

Source: Interviews with engineers at 10 robotics companies (March 2026)

High Cost
Outsourced digital twin **$1M+**

- 3D environment build outsourcing costs
- 3D asset purchasing costs

Wasted Time
Outsourcing communication **2hrs/day**

- Scene setup and variation work takes days

Poor Training Performance
Useless a 'Pretty 3D map'

- Doesn't reflect real conditions (weather, surfaces)
- Hard to replicate factory/field-level sim fidelity
- Sim gains don't carry over to reality

High Barrier to Entry
Isaac Sim — powerful, but nobody actually uses it

- Most developers have never touched it
- High learning curve (training, time, expertise)
- No accessible UI for non-experts

Solution

Sequor Robotics SW product "Real 3D Gym" — lowering the barrier to simulation adoption

Key Features 1

Scan customer's space with a camera to create the learning-friendly environment

Key Features 2

Custom robot / sensor plugin

Key Features 3

Full end-to-end features for robot learning
(tasks, training methods, AI models)

Real 3D Gym Workflow

Simplifying a complex, manual-heavy workflow

STEP 01
Real

Collect real-world spatial data

Scan the physical space using cameras

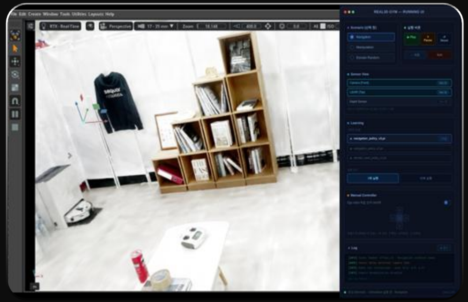
STEP 02
Real

Convert real space into a 3D virtual environment

Generate a 3D space ready for robot learning

STEP 03
Simulation

Set up the Real3D Gym environment



Set up learning-ready environment directly in the simulator

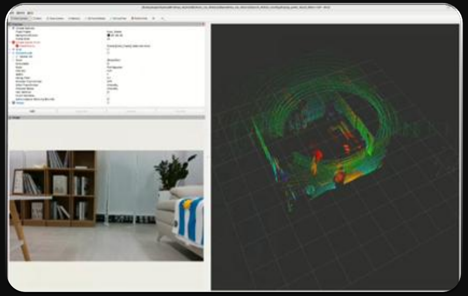
STEP 04
Simulation

Iterative training in a virtual environment

Optimize robot behavior through thousands of simulations

STEP 05
Real

Deploy learned policies to real robots



Transfer virtual training results to the actual robot

Future Use Case

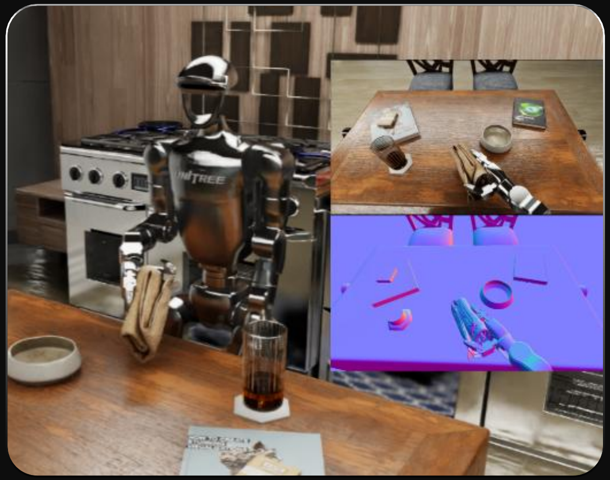
Sequor's software
— expanding what humanoids can do

Home — Cooking Assistant

Assisting with meal preparation and cleanup in kitchens and dining areas.

1X NEO

home humanoid performing household chores



Scenario Flow

S1	S2	S3	S4	S5	S6
Retrieving ingredients from the fridge	Preparing necessary utensils	Clearing the dining table	Loading the dishwasher	Wiping the table	Kitchen cleanup

Task

Total ~160 Tasks

Navigation	Pick & Place	Open	Close
20 TASK	80 TASK	30 TASK	30 TASK
Moving to the fridge and dining table; obstacle avoidance	Picking up ingredients, arranging utensils, clearing and organizing dishes	Opening the fridge, cabinets, drawers, and trash cans	Closing the fridge, cabinets, drawers, and trash cans

Deliverables for Robotics Clients

World Files	3D Assets: Furniture	3D Assets: Food
100 Environments	Various sizes	150+ Items
Home environment scans	Kitchen furniture assets	Ingredients and cooking tools

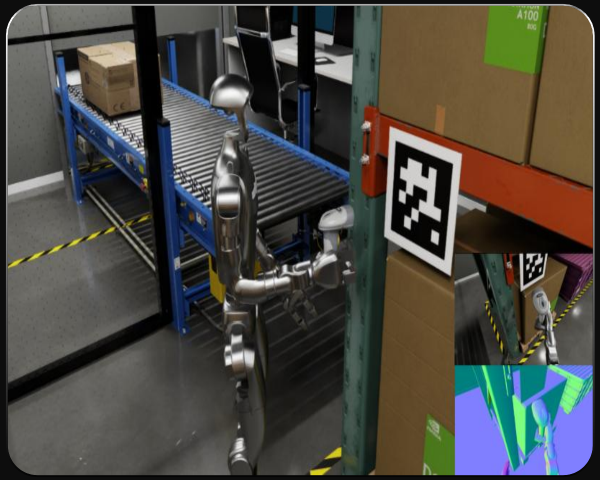
Future Use Case

Sequor's software
— expanding what humanoids can do

Apptronik Apollo
built for logistics & manufacturing

Logistics DAS — SKU Picking & Packing by Order

Shelf picking → Order bin loading → Box packing → Conveyor transfer



Scenario Flow

S1	S2	S3	S4	S5	S6
Receive order info	Move to shelf & search SKU	Pick item	Load into order bin	Pack box	Transfer to conveyor

Task

Total ~200 Tasks

Navigation	Pick & Place	Sorting	Packing & Transfer	Cleanup
15 TASK	100 TASK	40 TASK	25 TASK	20 TASK
Move between shelves and workstations	SKU retrieval & precise placement	Order bin sorting, mislabel prevention	Box packing, conveyor transfer	Box assembly, taping, returns

Deliverables for Robotics Clients

World File	Structures	SKU Products	Packaging
1 environment	15 items	100 SKUs	20 items
DAS warehouse scan	Shelves, bins, conveyors	Product packages	Boxes, raw materials

Sequor's Key Advantage — Time Savings

From multi-team handoffs → one person, one platform

7 sequential steps

BEFORE

Required Personnel: 6+ teams

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
Site survey & drawing acquisition	3D environment modeling	Simulator setup	Robot model setup	Training design & execution	Results review & debugging	Real robot deployment
Facilities team	Graphics outsourcing	Platform team	HW team	ML·RLteam	QA team	Field team

Any error means starting the previous step over

1 platform

AFTER

Required Personnel: **1 person**

Sequor Robotics	Sequor Robotics	Customer	Customer
Site Scan	Environment File Delivery	RL Training	Robot Deployment

Accelerate robot development by shortening robot learning cycles

Sequor's Key Advantage — Lower Barrier to Entry

Making Complex Robot Development Easier

Existing Approach

High Dependence on Specialized Personnel

QA/QC Engineer

Deployment Engineer

Robot Engineer

Simulation Engineer

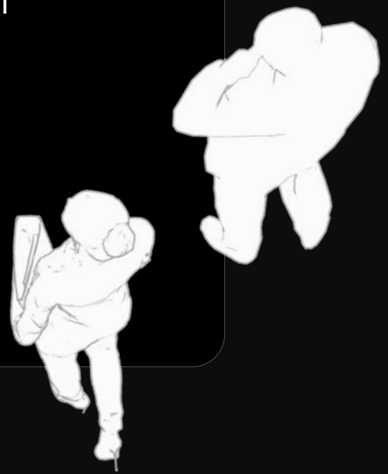
RL Engineer

Sequor Robotics

Robot Training for Everyone

Even non-experts can use it

Intuitive guidelines & UI
Consulting support for
Isaac Sim adoption

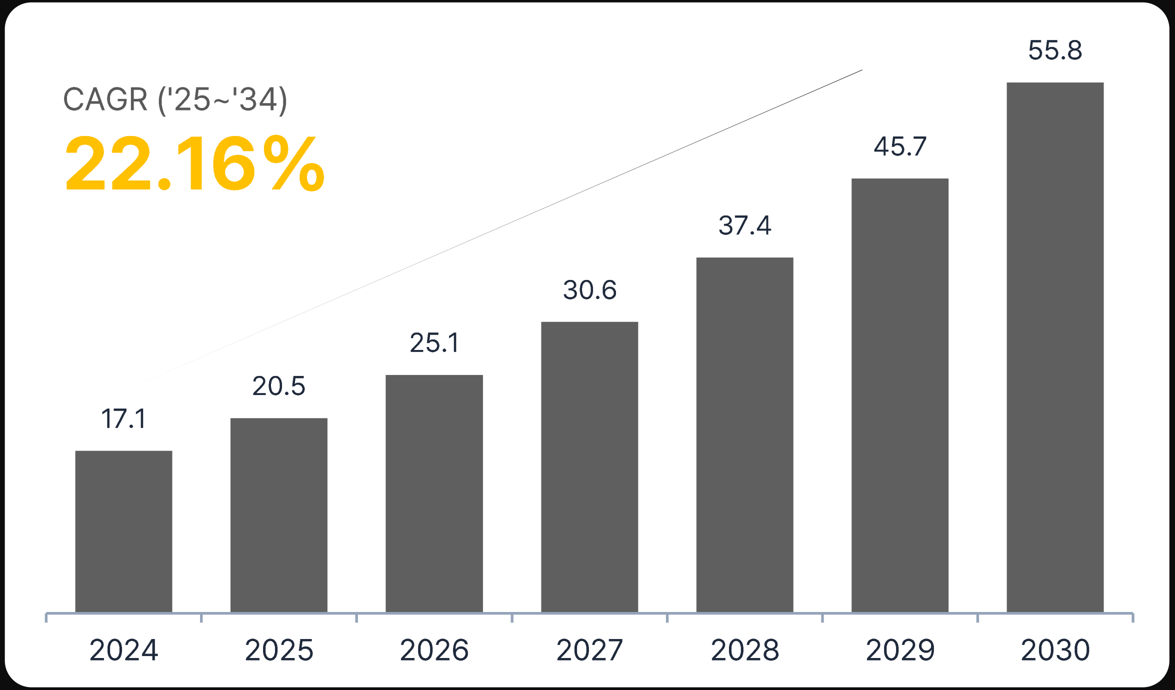


A platform that lowers the barrier to simulation adoption
by eliminating complex multi-team workflows.

The AI robot training market is exploding — simulation is at the core.

AI in Robotics Market Size

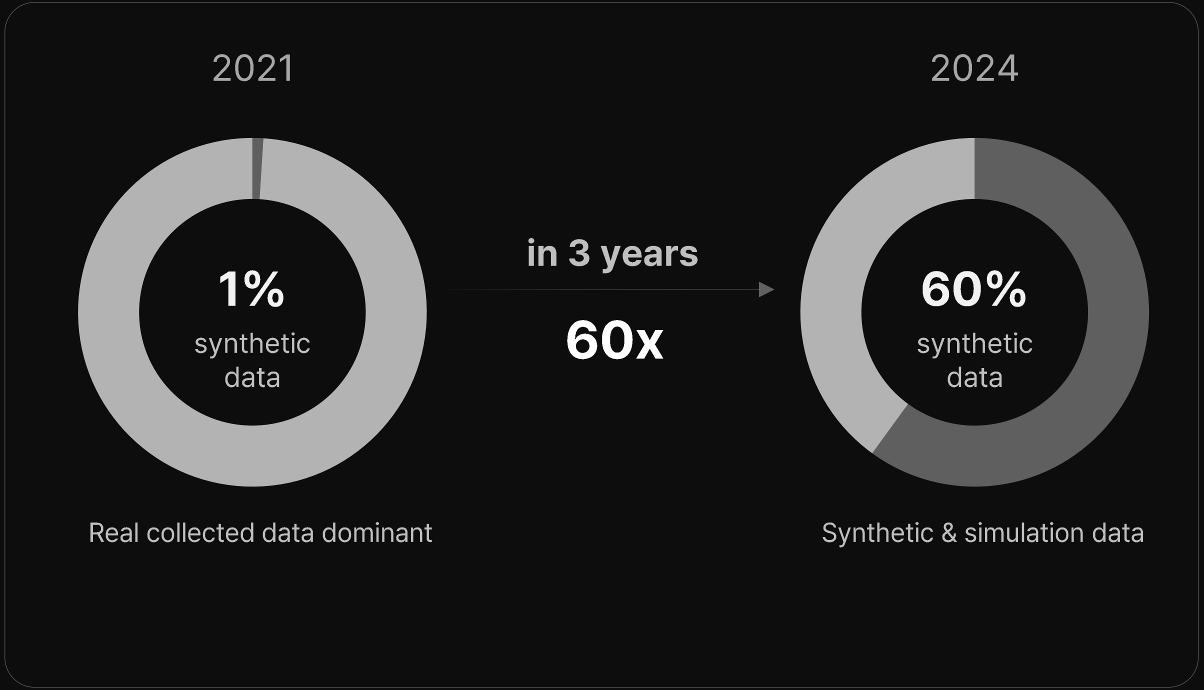
(Unit: USD Billion)



*Source: Precedence Research, 2025

A Paradigm Shift in AI Training Data

Synthetic & simulation-based data is becoming the dominant source of training data



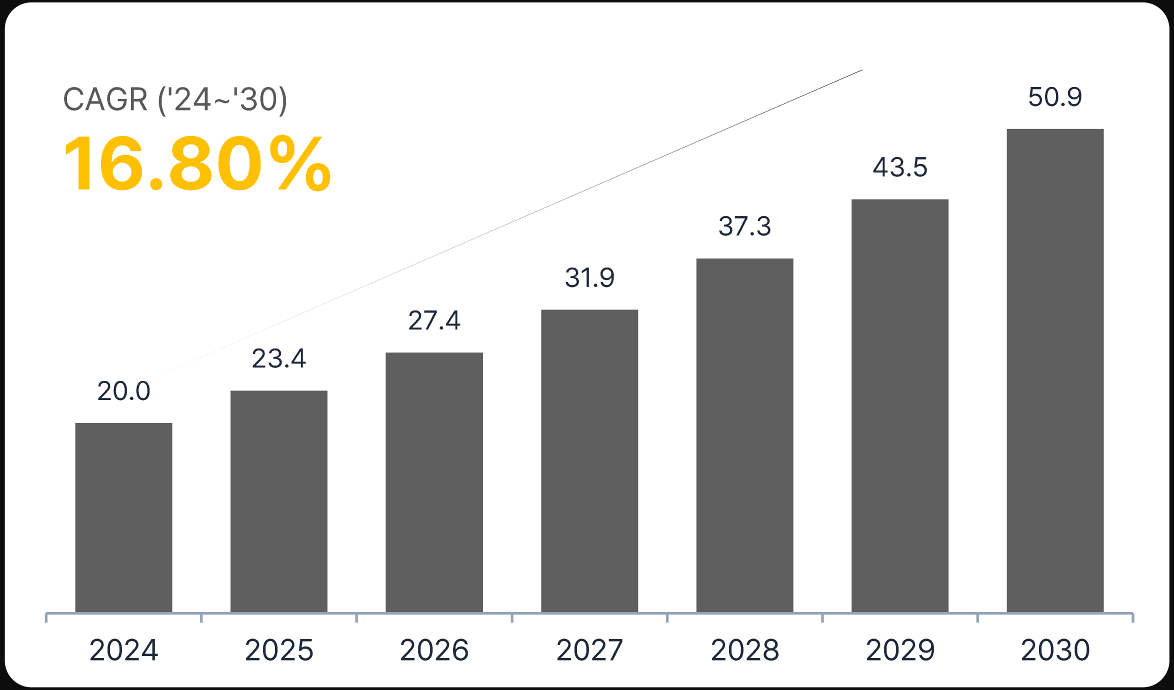
*Source: Gartner

Market Outlook

The robot simulation market is shifting toward software

Robot Simulation Market Size

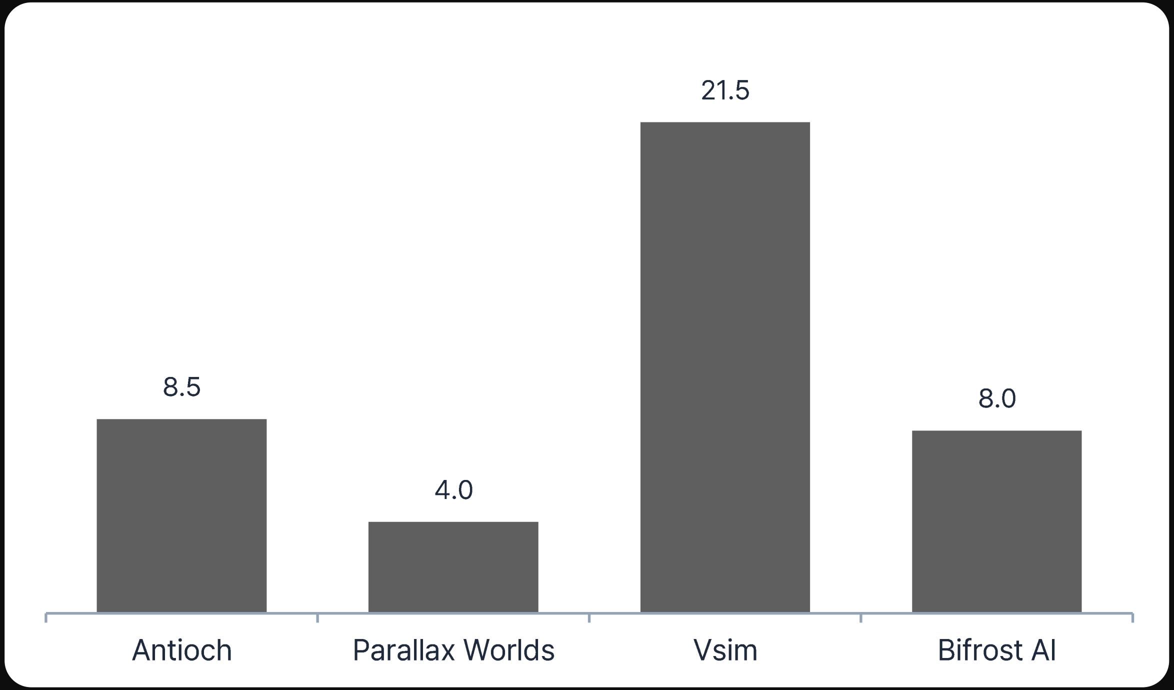
(Unit: USD Billion)



*Source: Cognitive Market Research, 2024

Investment Trends in the Robot Simulation Market

(Unit: USD Million)



Vision

A world where robots act on their own with just a single word

Now	Mid-term Goal	Long-term Goal
Real 3D Gym Beta	**3D World Model**	**Robot Foundation Model**
A simulation environment where robots can learn	**A spatial intelligence model that understands the real world.**	**Intelligence that enables robots to make decisions autonomously**
Creates realistic 3D environments based on homes, factories, and offices where robots can learn.	Understands real-world spaces, structures, and objects for physical interaction.	Allows robots to understand situations, plan, and act on their own.

Sequor's Technical Edge

30 Papers

- 2023–2026
- Robotics, computer vision, and AI
- Top international conferences and journals (NeurIPS, CVPR, ICRA, etc.)

4 Patents

Technologies that reduce the Sim-to-Real Gap

3D Spatial Structure Generation

Core Tech 1 — helps robots better understand simulation

Sensor Input

Core Tech 2 — helps robots better understand simulation

Training Efficiency Research & Validation

Research to improve robot training efficiency in simulation

Real-time robot perception software

3D World Model

Tech for helping robots better understand real world

Robot Foundation Model

Sequor's Technology 1

3D spatial structure optimized for robot learning

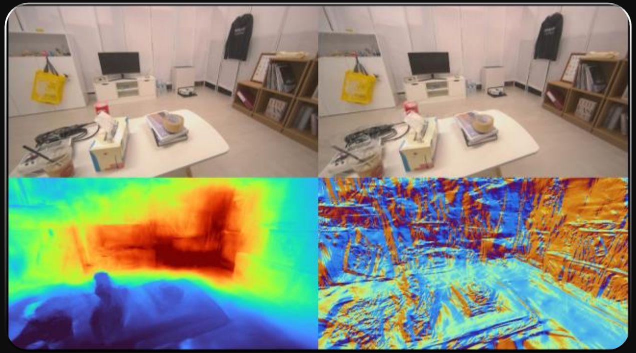
Robots don't just need visually appealing environments — **collision accuracy, distance, and spatial structure** must be precise for high-performance learning.

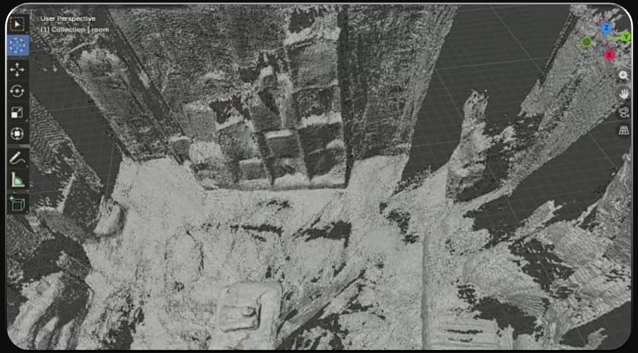
Comparison of learning suitability across commonly used environment tools

Method	Collision Geometry (robot collision & distance accuracy)	Visual Quality (real-world fidelity)	Robot learning Suitability
Visual SLAM (Company A)	✅ Strong	❌ Weak	Partially suitable
Neural Rendering (Company B)	❌ Weak	✅ Strong	Partially suitable
Sequor Real 3D Gym	✅ Strong	✅ Strong	✅ Suitable

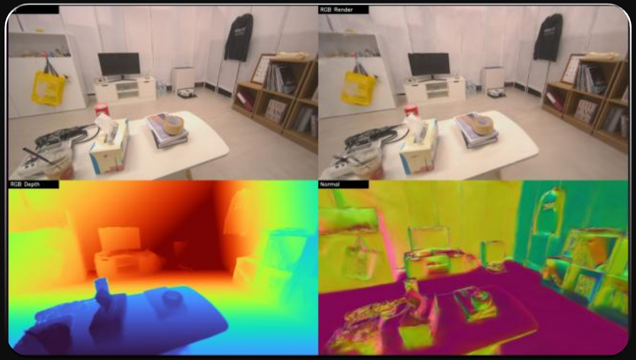
Sequor has developed 3D spatial generation technology that satisfies both visual quality and physical structure for robot learning.

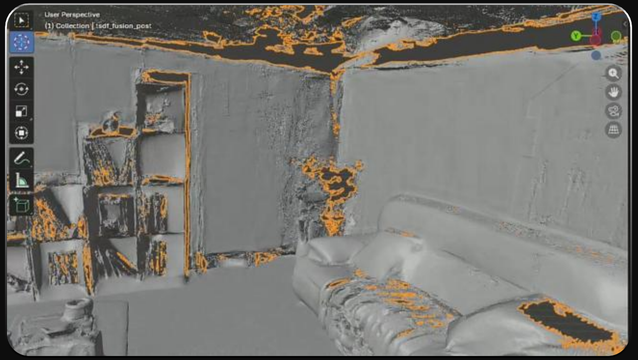
Company B Visual quality-focused environment reconstruction





Sequor Robotics Accurate spatial structure reconstruction optimized for robot learning



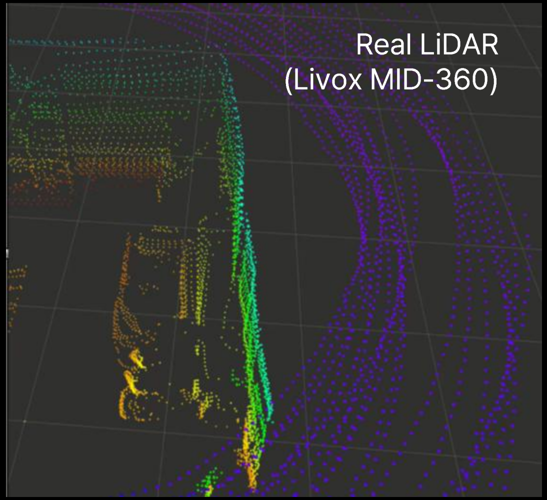


Accurate 3D structure directly impacts sim-to-real performance and learning stability

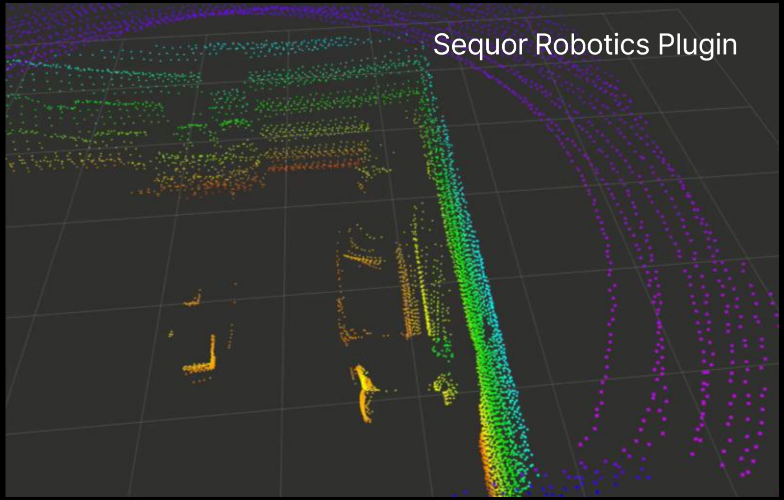
Sequor's Technology 2

Sensor data that replicates real-world conditions

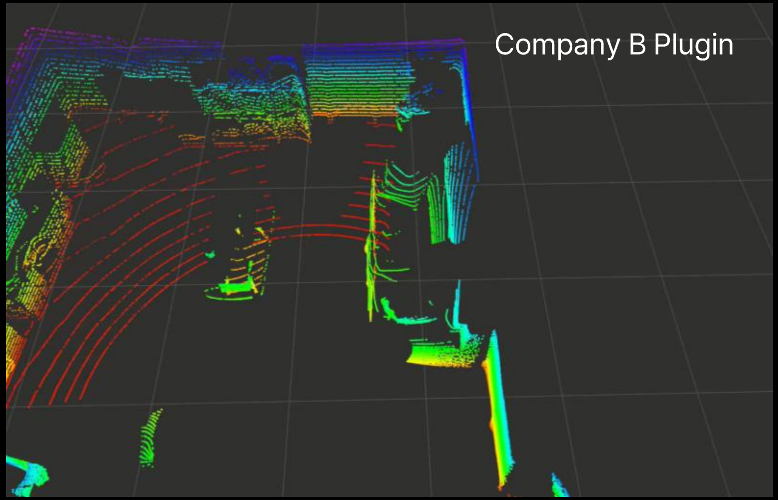
Many robot engineers say the gap between simulation and real sensor data requires constant extra work.



Sensor input gap is a main cause of sim-to-real performance degradation



Replicates real LiDAR distribution



Default sensor model

Sequor has built input data that closely replicates real sensor characteristics to improve training performance and deployment stability.

Sequor's Technology 3

Simulation learning that transfers reliably to the real world

Sequor has directly validated through research that robot policies learned in simulation perform stably in real-world environments.
Combining real and simulation data improves training performance (Published at IROS 2025)

Learning Data	Success Rate
Real	30%
Simulation	60%
Hybrid (Real + Simulation)	**75%**

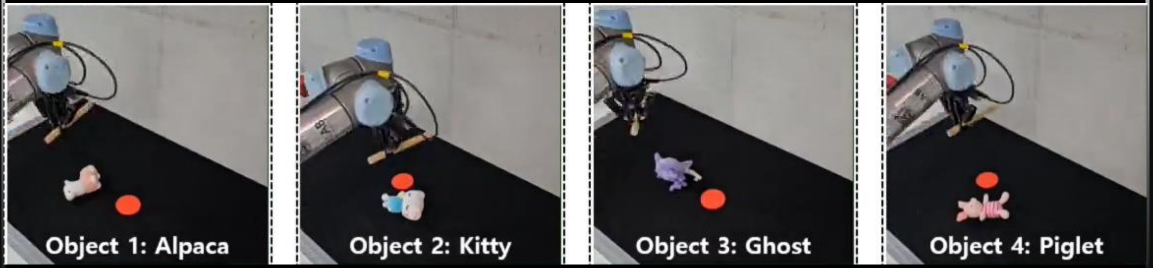
Navigation Task

Environment	Push to target	Wiping	Stick pushing
Simulation	96.3%	91.2%	86.9%
Real	92.5%	87.5%	85%
Gap	**3.8%**	**3.7%**	**1.9%**

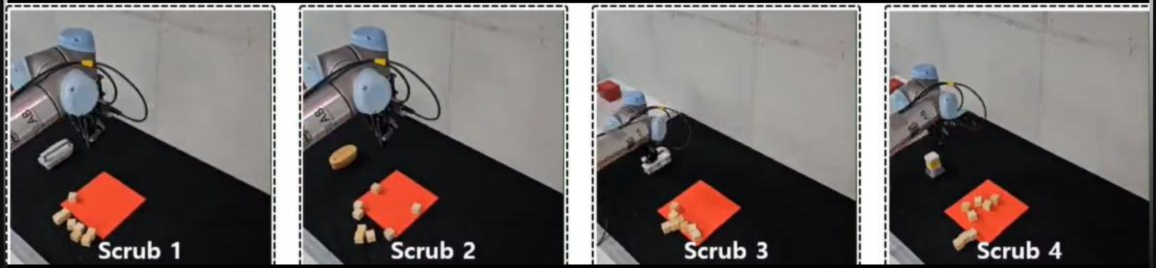
Simulation data alone achieves high success in real-world

Manipulation Task

Push to target



Wiping



Sequor's Technology **4**

3D World Model

An AI model that recognizes space in real time using only a single camera and generates a 3D map — a core foundation for Robot Foundation Models.

Existing 3D AI struggled with real-time processing because spatial data is extremely large.

Sequor solves this by dramatically reducing computation, enabling real-time operation even on low-power embedded boards.

3D Mapping

Precisely reconstructs indoor spaces in 3D using only a single camera — no LiDAR required.

Visual Localization

Instant localization from a single image

98%+
Localization accuracy

Real-Time Acceleration

1.5 FPS → 4.7 FPS achieved
Targeting 10 FPS for commercialization

Continuous Pipeline

Automated Cycle

Learning · Deployment · Data Collection · Retraining

Space Scan
LiDAR · RGB-D

3D Environment
Reconstructed 3D space

Simulation
Real 3D Gym
Simulation Learning

Real world
Real-World Deployment
Policy applied to hardware

Real world
Data Collection
Deployment environment data

Real world
Foundation Model
Model update

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A world where robots act on their own with just a single word

Milestones

2026 H2

Real 3D Gym beta launch & domestic PoC secured
U.S. exhibitions and accelerators → U.S. customer pipeline established

- IBK Changong
- Physical AI Superstart

Begin partnerships with U.S. companies
(In partnership discussions with a real data company)

2027 H1

Real 3D Gym product launch
- Commercialization of high-demand technologies (Automated environment setup & scenario generation)

PoC to Paid Conversion
Securing Domestic Clients

2027 H2

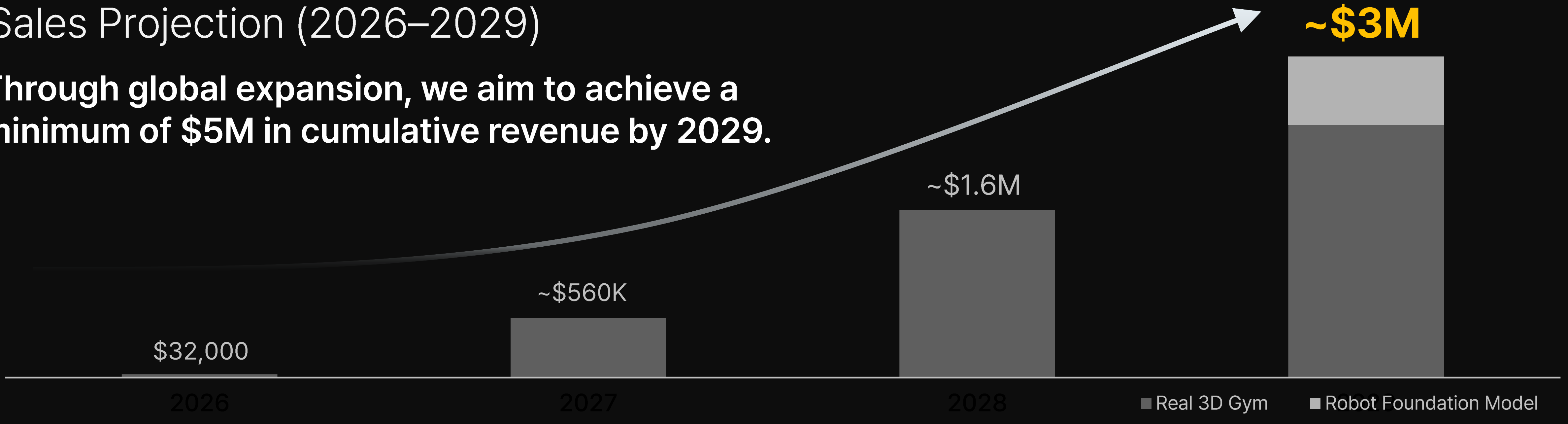
U.S. entity incorporated & initial local team in place
U.S. paying customers secured

2028 H1

3D-VLA software product launch
Series B fundraise

Sales Projection (2026–2029)

Through global expansion, we aim to achieve a minimum of $5M in cumulative revenue by 2029.



2026

Real 3D Gym

- Beta v1.0 launch
- Domestic paid PoC: 3 clients (LG CNS, ioCrops, etc.)
- Environment scans across logistics, manufacturing, and service sectors

2027

Real 3D Gym

- (H1) 3 U.S. reference clients secured
  → 2 scans + 1 task pipeline
- (H2) 4 domestic reference clients
  → scan → task full-stack conversion

2028

Real 3D Gym

- Full entry into U.S. market
- 10+ cumulative global clients
- First commercial humanoid reference

Robot Foundation Model

- 3D-VLA software complete
- Pilot client selection underway

2029

Real 3D Gym

- 20+ cumulative global clients
- Platform productization
  → Fully remote, globally deliverable structure established

Robot Foundation Model

- Foundation Model launch
- 3D Gym clients convert to subscription
- Full vertical subscription launch

TEAM

Building products and conducting research with a team of specialists across multiple fields

Head of Business, CEO & Co-Founder
Jeongwoo Oh

Ph.D. in Robot Learning, Seoul National Univ.
B.S. in Electrical and Computer Engineering, Seoul National Univ.
Forbes 30 Under 30, Korea/Asia

Head of Technology, CTO & Co-Founder
Songhwai Oh

Professor, Robot Learning Lab, Seoul National Univ.
Ph.D. in Electrical Engineering & Computer Science, UC Berkeley
Former Senior Engineer, Intel & Synopsys

Product Lead
Expertise: Robot Application
#SNU M.S. #CLOBOT #Siemens

AI/Robotics Researcher
Expertise: Latent 3D World Model, Neural 3D Map
#SNU Ph.D.

AI/Robotics Researcher
Expertise: Real-to-Sim-to-Real
#SNU Ph.D.

3D Vision Researcher
Expertise: 3D reconstruction, Vision AI
#Camera Hardware Specialist #KU B.S.

Company History

2022
Dec: Sequor Robotics incorporated

2023
Mar: Seed investment from Future Play
Jul: Selected as a Deeptech Tips company

2024
Mar: Selected for DIPS 1000+
Jul: Selected for Startup Autobahn Korea (LG Electronics partnership program)
Aug: Seed investment from BA Partners
Nov: Selected for LG SuperStartup Cohort 3
Nov: NDA signed with Samsung Electronics

2025
Apr: NDA signed with Samsung Electronics America
May: Patent filed — 3D spatial semantic occupancy prediction
Jun: Selected for IBK Changong Accelerating Program
Jul: PoC contract signed with LG Electronics
Aug: MOU signed with Samsung Electronics; registered as Samsung Electronics supplier
Aug: Patent filed — cleaning path optimization
Sep: Patent filed — 3D map management
Nov: Participated in Samsung Electronics AI Alliance
Dec: Patent filed — bird's-eye-view-based robot path planning
(Samsung Electronics, Namuga, Pixelplus, Sequor Robotics)
Nov: Selected for Deeptech Value Program (LG Electronics HS Division)

2026
Jan: Patent registered — 3D spatial semantic occupancy prediction
May: Patent registered — sparse octree-based semantic voxel inference

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Scaling Physical AI through Robot Foundation Models

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